

CLI Requirements

You agent must have a CLI that allows the orchestrator agent to call it as a subprocess. The CLI should accept user instructions and a layout JSON string as input, and return the agent's response as output along with a JSON string representing the edited layout if applicable.

Instructions for updating `main.py` to support CLI

1. In `team_XX/python/main.py`, switch from a single positional argument to explicit flags:

- `--prompt` (required string)
- `--layout_json` (optional JSON string)

Update any existing code that references `args[0]` to use `args.prompt` instead.

2. Import `json` and parse `--layout_json` when provided:

- Use `json.loads(args.layout_json)`.
- If parsing fails, raise a clear `ValueError` (or print an error and exit with non-zero code).

3. Initialize context with `ctx = bootstrap()` as before, then override layout context when CLI input is provided:

- If `--layout_json` is present, set `ctx.layout_data` to the parsed dict before calling `run_agent(...)`.
- This ensures the run uses orchestrator-provided layout data instead of only `layout_input/layout_schema.json`.

4. Execute the graph with the parsed prompt:

- `response = run_agent(prompt_text, ctx)`
Keep the output safe for terminals.

5. Print output using a stable machine-readable structure for orchestrators:

Final Response:

<agent response>

Edited Layout JSON:

<edited layout JSON or "No layout changes">

Example usage:

```
python main.py --prompt "add a window to the south wall of the living room" --layout_json '{
  "layoutId": "Layout-101",
  "outline": [[0.0, 0.0], [9.0, 0.0], [9.0, 5.0], [0.0, 5.0], [0.0, 0.0]],
  "rooms": [
    {
      "id": "room-1",
      "name": "Living Room",
      "geometry": [[0.0, 0.0], [5.0, 0.0], [5.0, 5.0], [0.0, 5.0], [0.0, 0.0]],
      "attributes": {"area": 25.0}
    },
    {
      "id": "room-2",
      "name": "Bedroom 1",
      "geometry": [[5.0, 0.0], [9.0, 0.0], [9.0, 5.0], [5.0, 5.0], [5.0, 0.0]],
      "attributes": {"area": 20.0}
    }
  ],
  "doors": [
    {
      "id": "door-1",
      "type": "wooden",
      "name": "Bedroom Door",
      "geometry": [[5.0, 2.0], [5.0, 2.9]],
      "attributes": {"connectsRooms": ["room-1", "room-2"]}
    }
  ],
  "windows": [
    {
      "id": "window-1",
      "type": "sliding",
      "name": "Living Room Window",
      "geometry": [[0.0, 2.0], [0.0, 3.5]],
      "attributes": {"roomId": "room-1"}
    }
  ],
  "furniture": [
    {
      "id": "furn-1",
      "name": "Main Couch",
      "geometry": [[2.0, 3.0], [4.0, 3.0], [4.0, 4.0], [2.0, 4.0], [2.0, 3.0]],
      "attributes": {"roomId": "room-1"}
    }
  ]
}
```

```

],
  "mep": [
    {
      "id": "mep-1",
      "name": "Living Room AC",
      "geometry": [[2.5, 4.5], [3.5, 4.5], [3.5, 4.8], [2.5, 4.8], [2.5, 4.5]],
      "attributes": {"system": "hvac"}
    }
  ],
  "structure": [
    {
      "id": "wall-1",
      "name": "North Interior Wall",
      "geometry": [[5.0, 0.0], [5.0, 5.0]],
      "attributes": {}
    }
  ]
}'

```

The agent should parse the `--prompt` and `--layout_json` arguments, run the agent graph, and print the final response and edited layout JSON to the console.

If there are any follow-up questions or clarifications needed, the agent should prompt the user for input in the console if launched from the CLI. This will allow the orchestrator agent to have a back-and-forth conversation with your agent if needed, and allow the user to provide additional information or clarification as the agent is running.

Example output:

Final Response:

"Added a window to the south wall of the living room."

Edited Layout JSON:

```
{
  "layoutId": "Layout-101",

  "outline": [[0.0, 0.0], [9.0, 0.0], [9.0, 5.0], [0.0, 5.0], [0.0, 0.0]],

  "rooms": [
    {
      "id": "room-1",
      "name": "Living Room",
      "geometry": [[0.0, 0.0], [5.0, 0.0], [5.0, 5.0], [0.0, 5.0], [0.0, 0.0]],
      "attributes": {
        "area": 25.0
      }
    },
    {
      "id": "room-2",
      "name": "Bedroom 1",
      "geometry": [[5.0, 0.0], [9.0, 0.0], [9.0, 5.0], [5.0, 5.0], [5.0, 0.0]],
      "attributes": {
        "area": 20.0
      }
    }
  ],

  "doors": [
    {
      "id": "door-1",
      "type": "wooden",
      "name": "Bedroom Door",
      "geometry": [[5.0, 2.0], [5.0, 2.9]],
      "attributes": {
        "connectsRooms": ["room-1", "room-2"]
      }
    },
    {
      "id": "door-2",
      "type": "wooden",
      "name": "Living Room Door",
```

```
    "geometry": [[5.0, 2.0], [5.0, 2.9]],
    "attributes": {
      "connectsRooms": ["room-1", "room-2"]
    }
  }
],
```

```
"windows": [
  {
    "id": "window-1",
    "type": "sliding",
    "name": "Living Room Window",
    "geometry": [[0.0, 2.0], [0.0, 3.5]],
    "attributes": {
      "roomId": "room-1"
    }
  },
  {
    "id": "window-2",
    "type": "sliding",
    "name": "Living Room South Window",
    "geometry": [[2.0, 0.0], [3.5, 0.0]],
    "attributes": {
      "roomId": "room-1"
    }
  }
],
```

```
"furniture": [
  {
    "id": "furn-1",
    "name": "Main Couch",
    "geometry": [[2.0, 3.0], [4.0, 3.0], [4.0, 4.0], [2.0, 4.0], [2.0, 3.0]],
    "attributes": {
      "roomId": "room-1"
    }
  }
],
```

```
"mep": [
  {
    "id": "mep-1",
```

```
    "name": "Living Room AC",
    "geometry": [[2.5, 4.5], [3.5, 4.5], [3.5, 4.8], [2.5, 4.8], [2.5, 4.5]],
    "attributes": {
      "system": "hvac"
    }
  },
  {
    "id": "mep-2",
    "name": "Main Breaker Box",
    "geometry": [[0.2, 0.2], [0.8, 0.2], [0.8, 0.5], [0.2, 0.5], [0.2, 0.2]],
    "attributes": {
      "system": "electrical"
    }
  }
],

"structure": [
  {
    "id": "wall-1",
    "name": "North Interior Wall",
    "geometry": [[5.0, 0.0], [5.0, 5.0]],
    "attributes": {}
  }
]
}
```